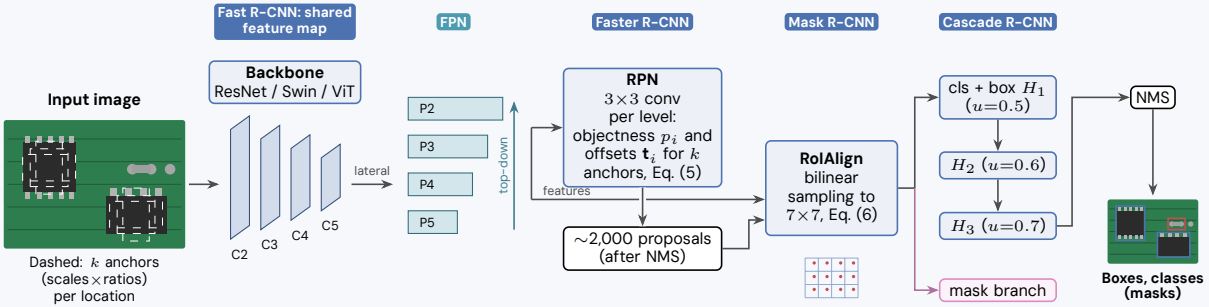
Sec. 3.1 Generalised two-stage detector (Faster R-CNN family)



Training. RPN anchors with IoU≥ 0.7 are positives and those with IoU≤ 0.3 negatives; proposals with IoU$\geq u$ train each head. The loss $\mathcal{L} = \mathcal{L}_{\text{RPN}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{box}}$ uses smooth-L_1 or IoU-based box terms (Eqs. (9)–(10)).

Variants. R-FCN replaces the per-RoI head with position-sensitive score maps; Soft-NMS decays rather than removes overlapping boxes.